Kinematics of deformable continuous bodies

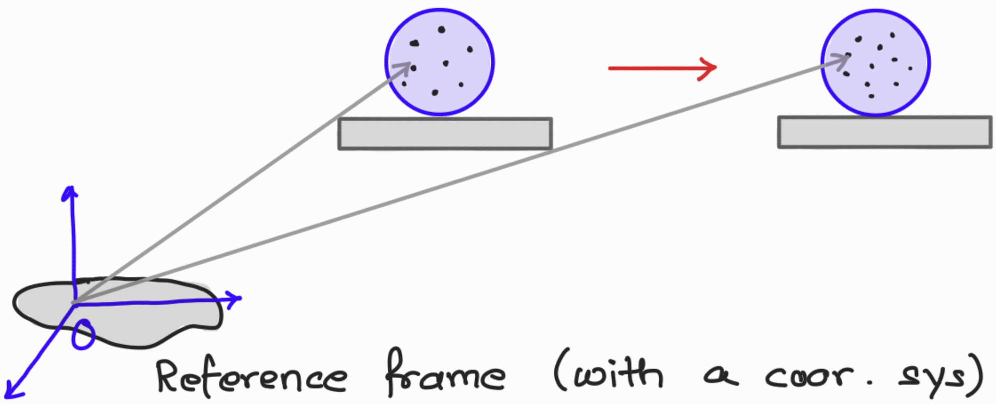
As of now, we have discussed the stress tensor and its various properties.

Now we will delve into the kinematics of a deformable body

Kinematics deals with expressing the deformation and motion of material bodies in mathematical form

deformation gradient, strain, etc. which allow to describe a particular aspect of a deforming material

No consideration is given to what is causing the deformation and movement (the cause being the action of forces/loads)

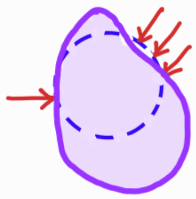


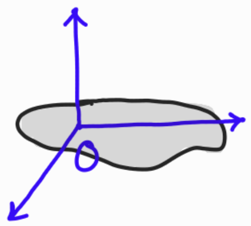
A physical ball (enclosing a set of material points in space) is placed on a slab and the slab is moved

Each particle of the ball undergo a displacement

The particles of the ball have moved in space as a rigid body. The material of the ball remain unstressed

On the other hand, when the same ball is acted upon by a set of forces, it changes its shape and/or size, that is, it deforms



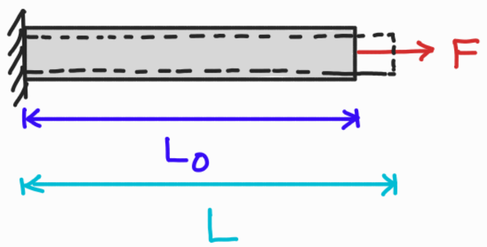


This deformation can be described using the concept of strain

Strain

One-dimensional strain — Engineering strain

Think of a horizontal bar subjected to a force F



If L_0 is the original length and L is the final length, we defined (in school days) that strain in the bar as

$$\text{Strain} = \frac{\text{Change in length}}{\text{Original length}},$$

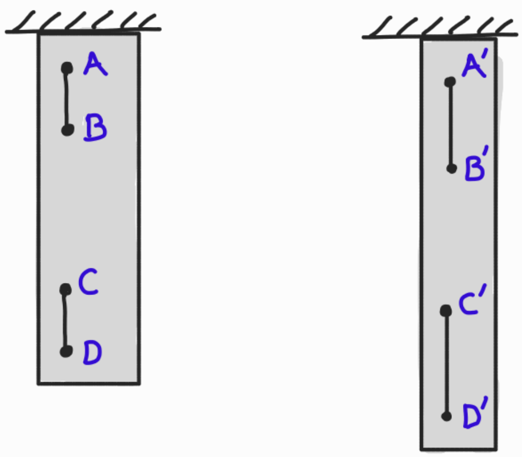
Engineering strain

$$\boxed{\epsilon = \frac{L - L_0}{L_0}}$$

However, the above definition is not useful for many cases where the strain at different points are different.

e.g. A bar hanging vertically under its own weight

In this case, if we consider two small line elements at two different locations: one near clamped end and one near free end, then the latter undergoes a much larger change in length. Thus, the strain in a body can vary from point to point.



$$\epsilon^{(A)} = \frac{\|A'B'\| - \|AB\|}{\|AB\|}$$

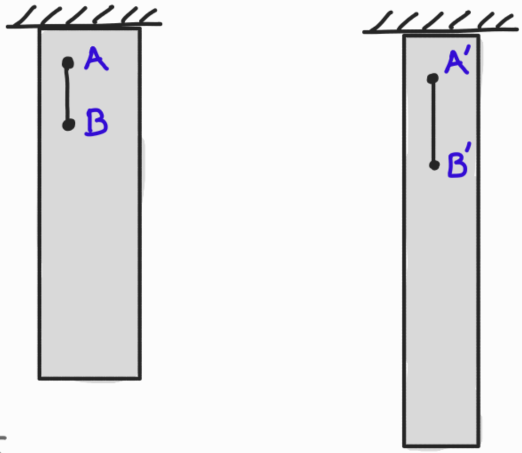
$$\epsilon^{(C)} = \frac{\|C'D'\| - \|CD\|}{\|CD\|}$$

When the strain is non-uniform, e.g. $\epsilon^{(A)} \neq \epsilon^{(C)}$, then the earlier defn of strain using the length of the rod fails, and the length of the line element now DOES matter.

To be precise, to define strain at a pt, we should consider "infinitesimally small" line element at that pt.

Engineering strain at a point is thus defined by considering an infinitesimally small line segment at the pt of interest

$$\epsilon^{(A)} = \lim_{\|AB\| \to 0} \frac{\|A'B'\| - \|AB\|}{\|AB\|}$$



where $\|\cdot\|$ denotes the magnitude of the line segment

So if a line segment is

- stretched to twice its original length $\rightarrow$ $\epsilon = 1$
- unstretched $\rightarrow$ $\epsilon = 0$
- shortened to half its original length $\rightarrow$ $\epsilon = -0.5$

Strain is often expressed in percentage

$\epsilon = 1 \Rightarrow$ 100% strain

$\epsilon = 2 \Rightarrow$ 200% strain

Most engineering materials, such as metals and concrete, undergo extremely small strains in practical applications in the range of $10^{-6} - 10^{-2}$ %

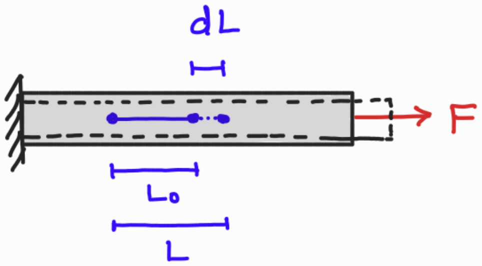
One-dimensional strain – True strain

There are many ways in which strain can be defined. Different strain measures are used in advanced theories of solid mechanics, particularly when deformations are large.

Apart from engineering strain, one other measure that is often used in material testing is called true strain (or logarithmic strain)

Define a small increment in strain to be the change in length divided by the current length

$$d\epsilon_t = \frac{dL}{L}$$



The total strain is the accumulation of these strain increments:

True strain → $$\epsilon_t = \int_{L_0}^{L} \frac{dL}{L} = \ln\left(\frac{L}{L_0}\right)$$ $\ln \rightarrow \log_e$

A usefulness of true strain is the fact that if a line element is stretched, true strain is +ve, while if the line element contracts (or shortens) then true strain is negative. For example, a line segment

- stretched to twice its original length $\epsilon_t = 0.69$
- unstretched $\epsilon_t = 0$
- shortened to half its original length $\epsilon_t = -0.69$

Another usefulness of true strain is that it is additive

For example, if a line element stretches in two steps from lengths L_1 to L_2 to L_3, the total true strain is

$$\epsilon_t = \ln\left(\frac{L_2}{L_1}\right) + \ln\left(\frac{L_3}{L_2}\right) = \ln\left(\frac{L_3}{L_1}\right)$$

which is the same as if the stretching had happened in one step. This is not the case with engineering strain.

The true strain and the engineering strain are related through

$$\epsilon = \frac{L - L_0}{L_0}$$

$$\epsilon_t = \ln(1+\epsilon)$$

$$\epsilon_t = \ln\left(\underbrace{\frac{L}{L_0} - 1}_{\epsilon} + 1\right)$$

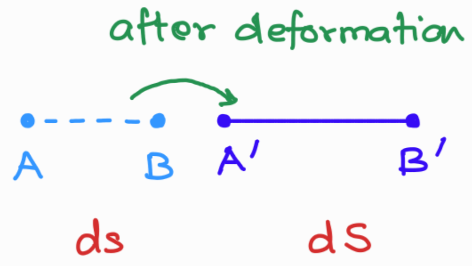
Smaller the deformation, lesser is the difference between the two strains

$$\epsilon_t = \ln(1+\epsilon) \approx \epsilon - \underbrace{\frac{\epsilon^2}{2}}_{\text{small}} + \underbrace{\frac{\epsilon^3}{3}}_{\text{small}} - \cdots \quad \text{for small } \epsilon$$

Almost all strain measures that are defined (for advanced theories) are similar in this way: they are defined s.t. they are more or less equal when the deformation is small

Put another way, when the deformations are small, it does not matter which strain measure is used, since they are all essentially very similar.

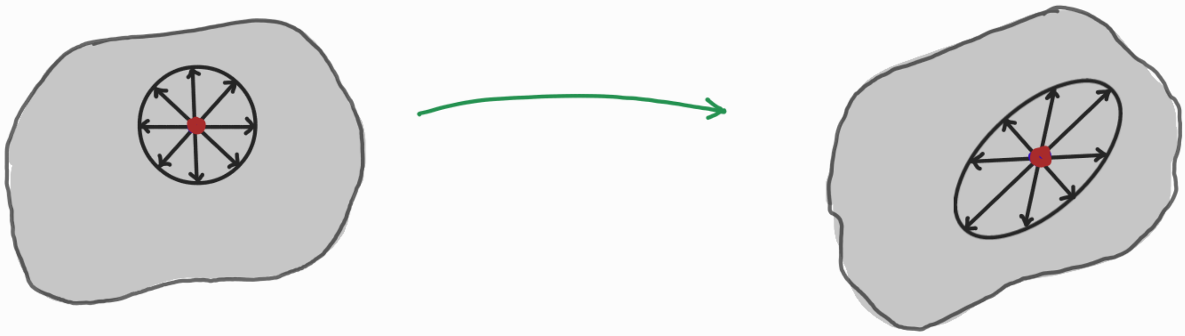
Apart from engineering and true strains, there are two other commonly used strain measures:



Strain measure	Description (in 1D)
1) Engineering strain	$\epsilon = \dfrac{dS - ds}{ds}$
2) True strain	$\epsilon_t = \ln(1+\epsilon)$
3) Green-Lagrange strain	$E_G = \dfrac{dS^2 - ds^2}{2\, ds^2}$
4) Euler-Almansi strain	$E_E = \dfrac{dS^2 - ds^2}{2\, dS^2}$

Concept of strain for a general 3D body

In an one-dimensional body, we can only talk about strain along only one dimension by considering a very small line element at a pt in the body aligned with the length of the body. However, for a general 3D body, one can consider many small lines at a point in different directions. For example, consider the deformation of line elements in a 2D deformed region



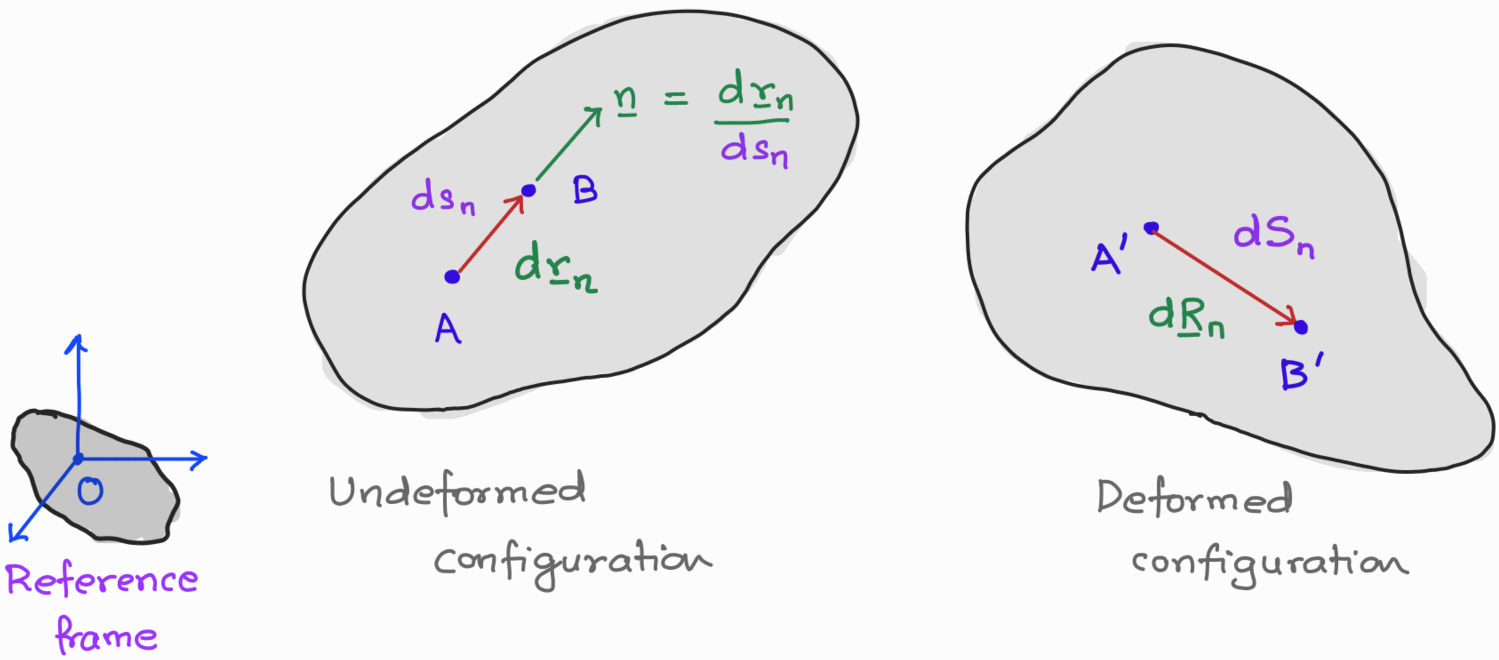
You can see the line elements along different directions at the same point experience different degrees of stretch, and not only that, they also undergo change in angle between the line elements during deformation.

We will next describe two types of strains that are used to characterize these changes at any pt within a body.

Normal Strain

When a body deforms, line elements inside the body may get elongated or shortened. In simple words, the quantity we use to measure the changes in length of a small line element at a point inside the body is called normal strain

Consider a small line element, $d\underline{r}_n$, directed along normal $\underline{n}$, between two close points A and B in an undeformed body. The length of $d\underline{r}_n$ being ds_n.



When the body deforms, point A moves to a new point A' and B moves to B' and the original line element will change to a new line segment A'B', with position vector $d\underline{R}_n$, of length dS_n

As the defn of strain is not unique, that is people have proposed equally valid definitions of strain, here will work with the Green-Lagrange strain measure (an exact measure) and define normal strain mathematically as:

$$E_{nn} = \frac{1}{2} \frac{dS_n^2 - ds_n^2}{ds_n^2}$$

$$= \frac{1}{2} \frac{d\underline{R}_n \cdot d\underline{R}_n - d\underline{r}_n \cdot d\underline{r}_n}{ds_n^2}$$

$$= \frac{1}{2} \left(\frac{d\underline{R}_n}{ds_n} \cdot \frac{d\underline{R}_n}{ds_n} - \underbrace{\frac{d\underline{r}_n}{ds_n}}_{\underline{n}} \cdot \underbrace{\frac{d\underline{r}_n}{ds_n}}_{\underline{n}} \right)$$

$$= \frac{1}{2} \left(\left\| \frac{d\underline{R}_n}{ds_n} \right\|_2^2 - 1 \right) \qquad \left[\because \ \|\underline{n}\| = 1 \right]$$

This definition of normal strain holds true regardless of the amount of strain — large or small. Therefore, this strain measure is an exact measure of strain.

Infinitesimal Normal Strain

If the change in the length of the line element is very small, say in the order of $10^{-6} - 10^{-3}$, the Green-Lagrange normal strain defn reduces to the Engineering strain :

$$E_{nn} = \frac{1}{2} \frac{dS_n^2 - ds_n^2}{ds_n^2} = \frac{1}{2} \frac{(dS_n - ds_n)(\overbrace{dS_n}^{\approx ds_n} + ds_n)}{ds_n^2}$$

Green-Lagrange normal strain (exact)

$$\approx \frac{1}{\cancel{2}} \frac{(dS_n - ds_n)(\cancel{2}\, \cancel{ds_n})}{ds_n^{\cancel{2}}}$$

Engineering normal strain → $\epsilon_{nn} = \dfrac{dS_n - ds_n}{ds_n}$

To define normal strain at the point A, we reduce the length ds_n of the line element s.t. in the limit of $ds_n \to 0$, we recover the normal strain at a point along the direction $\underline{n}$

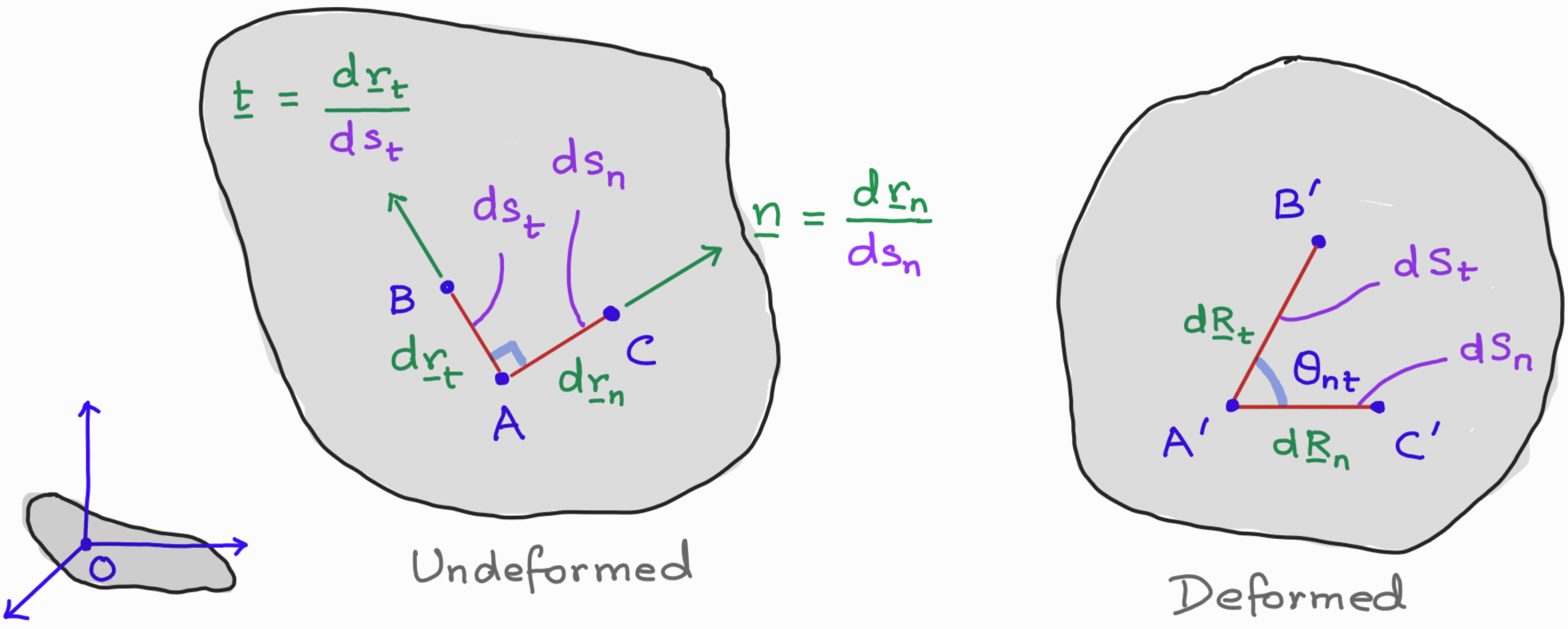
$$\therefore \quad \epsilon_{nn} = \lim_{ds_n \to 0} \frac{dS_n - ds_n}{ds_n}$$

Shear Strain

In addition to changes in length to line elements in a deformable body, there are also distortions that correspond to change in angle between two line elem.

The change in angle between two line elements which were initially perpendicular in the undeformed config. of the body is defined using shear strain

Consider two small line segments $d\underline{r}_n$ and $d\underline{r}_t$ along unit normal directions $\underline{n}$ and $\underline{t}$ respectively. The two unit normals $\underline{n}$ and $\underline{t}$ are chosen to be mutually perpendicular to each other in the undeformed config. of the body



In the deformed configuration, the line segments $d\underline{R}_n$ and $d\underline{R}_t$ may not longer be orthogonal. Let θ_{nt} be the angle between the two rotated line segments now. The Green-Lagrange shear strain (or the exact shear strain) is defined as

$$E_{nt} = \frac{1}{2} \frac{d\underline{R}_n}{ds_n} \cdot \frac{d\underline{R}_t}{ds_t} \quad \text{—— (1)}$$

We know that a dot product between two unit vectors yields the cosine of the angle between the two vectors

Therefore,

$$\boxed{\cos \theta_{nt}} = \frac{\frac{d\underline{R}_n}{ds_n} \cdot \frac{d\underline{R}_t}{ds_t}}{\left\| \frac{d\underline{R}_n}{ds_n} \right\| \left\| \frac{d\underline{R}_t}{ds_t} \right\|}$$

$$\sin\left(\frac{\pi}{2} - \theta_{nt}\right)$$

From our definitions of Green-Lagrange normal strain,

$$E_{nn} = \frac{1}{2}\left(\left\| \frac{d\underline{R}_n}{ds_n} \right\|^2 - 1\right) \Rightarrow \left\| \frac{d\underline{R}_n}{ds_n} \right\| = \sqrt{1 + 2E_{nn}} \quad \text{—— (2)}$$

Using rels 1 & 2 in the defn of $\sin\left(\frac{\pi}{2} - \theta_{nt}\right)$, we get

$$\sin\left(\frac{\pi}{2} - \theta_{nt}\right) = \frac{\frac{d\underline{R}_n}{ds_n} \cdot \frac{d\underline{R}_t}{ds_t}}{\left\| \frac{d\underline{R}_n}{ds_n} \right\| \left\| \frac{d\underline{R}_t}{ds_t} \right\|}$$

$$= \frac{2 E_{nt}}{\sqrt{1 + 2E_{nn}}\ \sqrt{1 + 2E_{tt}}}$$

- If $\theta_{nt} = \pi/2$, change of angle $= 0$ $\Rightarrow$ $E_{nt} = 0$

  $\Rightarrow$ no shear strain

- When the changes of lengths of line elements and the change in angle are very small, then

$$E_{nn} \ll 1, \quad E_{tt} \ll 1$$

$$\sin\left(\frac{\pi}{2} - \theta_{nt}\right) \approx \frac{\pi}{2} - \theta_{nt}$$

so that

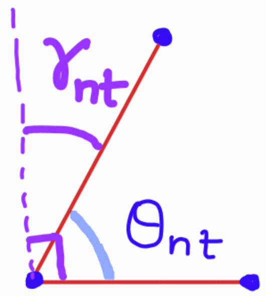
$$E_{nt} \approx \boxed{\epsilon_{nt} = \frac{1}{2}\left(\frac{\pi}{2} - \theta_{nt}\right)}$$

$\leftarrow$ infinitesimal shear strain

So for small strain case, we have infinitesimal shear strain defined at point in the body using very very small line elements along $\perp$ directions $\underline{n}$ and $\underline{t}$

$$\epsilon_{nt} = \frac{1}{2}\left(\frac{\pi}{2} - \theta_{nt}\right)$$

In engineering studies, it is common to use engineering shear strain γ_{nt}, in place of ϵ_{nt}, where



$$\gamma_{nt} \overset{def}{\equiv} 2\epsilon_{nt} = \frac{\pi}{2} - \theta_{nt}$$

The engineering shear strain is just equal to the change of angle so that it has a direct physical meaning. However, we will see that ϵ_{nt} is more convenient to use which will be a part of the strain tensor.